

Technical Writing

Lecture4

TEAMWORK AND COMMUNICATION

Teamwork

- Teamwork is essential in **engineering and software development**.
- Large software systems require **many developers working together**.
- Example: Developing a complex game like **Skryim** required a **large development team**.
- **Benefits of Teamwork**
 - **Team Synergy:** Teams produce better results than individuals working alone.
 - **Team Intelligence:** A team combines **different skills, knowledge, and ideas**.
 - Leads to **greater efficiency and innovation**.
- 💡 **Example for CS Students**
- Software projects often involve teams working on:
 - Programming
 - Testing
 - UI/UX design
 - Documentation
 - ⚠️ **Challenge:**
Conflicts may occur due to **poor communication or unclear goals**.



How to Make Teams Work Effectively

- **Strategies for Successful Teamwork**
- **Elect a Team Leader** – coordinates tasks and communication.
- **Define Clear Goals** – ensure all members understand the project objective.
- **Establish Team Rules** – set expectations for meetings, communication, and contribution.
- **Assign Responsibilities** – each member leads specific tasks.
- **Plan Meetings** – create agendas and record meeting minutes.
- **Schedule Tasks** – estimate time and use tools like **Gantt Charts**.
- **Manage Communication** – address problems early and focus on solutions.

Common Teamwork Tools

- **Common Teamwork Tools**
- **Team Charter** – defines team **rules, goals, and expectations**
- **Meeting Agenda** – lists **topics for discussion**
- **Meeting Minutes** – records **decisions and key points**
- **Work Logs** – tracks **tasks completed and time spent**
- **Status Reports** – shows **progress and remaining work**
- **Gantt Chart** – visual **timeline for tasks and deadlines**
-  **Example**
Used in **software development teams** to manage **projects, tasks, and deadlines.**

Five Models for Understanding Team Dynamics

- **Team Dynamics**
How team members **interact, communicate, and work together.**
- **Why it Matters**
- Improves **team performance**
- Helps manage **conflicts**
- Supports **better decision-making**
- **Team Dynamics Models Help Teams**
- Understand **different personalities**
- Improve **communication**
- Handle **problems or crises effectively**
-  **Example (Computer Science)**
In a **software development team**, understanding team dynamics helps coordinate **coding, testing, debugging, and documentation tasks.**

The Tuckman Team Model



The Tuckman Team Model

- **1. Forming**
- Team members **meet and get to know each other**
- Establish **team expectations and guidelines**
- Create a **Team Charter**
- Build **trust** among members
- **2. Storming**
- **Conflicts and disagreements** may occur
- Members share **different ideas and opinions**
- Team begins to identify **strengths, weaknesses, and roles**
- Learning to **manage conflict constructively** is important
-  **Example (CS Project):**
During brainstorming for a **software project**, team members propose different ideas for system design.

The Tuckman Team Model

- **3. Norming**
 - Roles and responsibilities become **clear**
 - Team members **cooperate and communicate effectively**
 - Work progresses smoothly
 - Tools like **Gantt Charts** help track progress
- **4. Performing**
 - Team works **efficiently and independently**
 - Strong **team synergy**
 - Tasks are completed **quickly and effectively**
- **5. Adjourning**
 - Project is **completed**
 - Team members **separate after finishing the work**
-  **Key Idea:**
Good **communication at every stage** helps teams move successfully to the next stage.

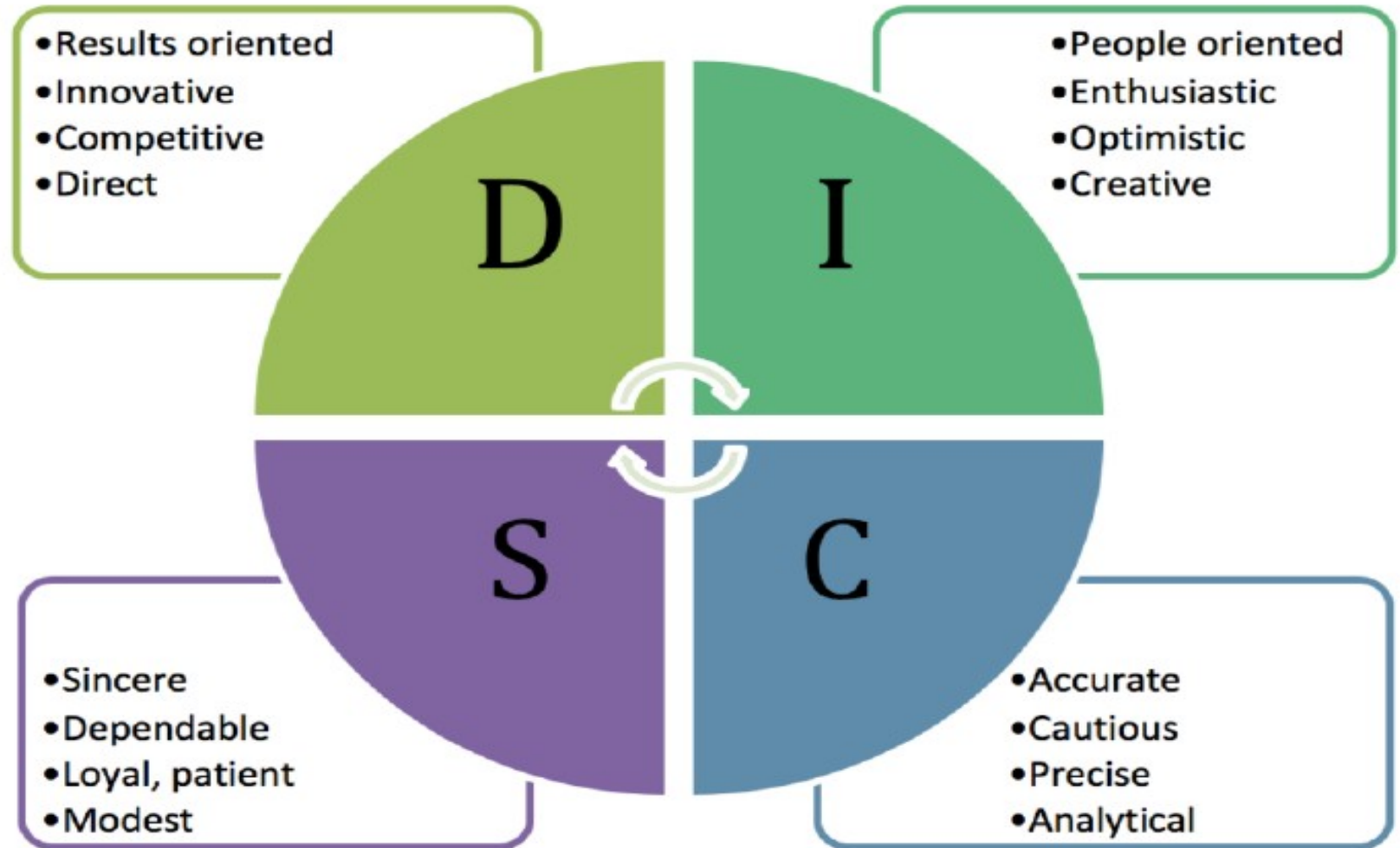
DISC Model

The **DISC model** helps teams understand **different personality styles** and improve collaboration.

Example for Computer Science Teams

- D**: Project leader or system architect
- I**: Presenter or team communicator
- S**: Team coordinator or tester
- C**: Programmer, analyst, or documentation specialist

People behave differently when they work in a team. Some people like to lead, some like to talk and motivate others, some prefer stability, and some focus on details. The DISC model divides these behaviors into four personality styles



DISC Model

D – Dominance

These people like **results and leadership**. They focus on finishing tasks quickly and making decisions.

Example :

The student who says:

“Let’s divide the work and finish the project today.”

Characteristics

- Direct, decisive, risk-taker
- Likes challenges and solving problems

Strengths

- Strong **organizer and decision maker**

- Innovative and goal-focused

Weaknesses

- May be **aggressive, stubborn, or impatient**

I – Influential (Inspiring)

These people are **social, enthusiastic, and good communicators**.

They motivate the team and enjoy interaction.

Example:

The student who talks a lot, encourages others, and likes presenting.

Characteristics

- Enthusiastic, persuasive, optimistic
- Strong communication and social skills

Strengths

- Good **motivator and negotiator**
- Creative problem solver

Weaknesses

- May **ignore details** or focus more on popularity than results

DISC Model

C – Conscientious (Cautious)

These people focus on **accuracy, details, and quality**.

They like rules, analysis, and correct information.

Example:

The student who checks grammar, numbers, and references in the report.

Characteristics

- Analytical, precise, systematic
- Focuses on **accuracy and quality**

Strengths

- Strong **research and analysis skills**

- Provides realistic and logical solutions

Weaknesses

- May focus **too much on details**
- May avoid conflict or hesitate to decide

S – Steady (Supportive)

These people are **patient, supportive, and reliable**.

They prefer cooperation and stability.

Example:

The student who helps everyone and keeps the group working smoothly.

Characteristics

- Friendly, reliable, patient
- Good listener and team player

Strengths

- **Dependable and cooperative**
- Good at **resolving conflicts**

Weaknesses:

May **resist change**

Sensitive to criticism

Key Components of Effective Teamwork (GRIP Model)



Thomas-Kilmann Conflict Mode Model

“When people disagree in a team, they usually respond in one of five ways. The Thomas-Kilmann model explains these five conflict styles.”

The model is based on **two factors**:

- **Assertiveness** → how strongly you try to achieve your own goal
- **Cooperativeness** → how much you try to help others reach their goal



Thomas-Kilmann Conflict Mode Model

- This model shows **how people respond to conflict in a team.**
It is based on **two simple ideas**:
- **Assertiveness:** How much a person tries to support **their own idea.**
- **Cooperativeness:** How much a person tries to **support others' ideas.**
-  **Competing**
- Focus on **winning the argument**
- Person pushes strongly for their idea
- **Example:**
A team member insists their algorithm must be used.
- **Good for:** quick decisions
Risk: may create tension in the team

Thomas-Kilmann Conflict Mode Model

-  **Collaborating**
 - Team members **work together to find the best solution**
 - Everyone's ideas are discussed
 - **Example:**
The team compares two coding solutions and combines the best parts.
 - **Good for:** strong teamwork and better solutions
-  **Accommodating**
 - A person **agrees with others to keep peace**
 - **Example:**
A student accepts another design idea even if theirs was different.
 - **Good for:** maintaining harmony
 - **Risk:** good ideas may be lost

Thomas-Kilmann Conflict Mode Model

- **Avoiding**
 - The conflict is **ignored or postponed**
 - **Example:**
Team members avoid discussing a disagreement about project tasks.
 - **Good for:** small issues
Risk: problems may remain unsolved
- **Compromising**
 - Each person **gives up something**
 - The team chooses a **middle solution**
 - **Example:**
Two interface designs are merged into one simplified design.
 - **Good for:** quick agreement
Risk: solution may not be the best one

The Five Dysfunctions of a Team (Lencioni's Model)

1. Absence of Trust

Meaning:

Team members do not trust each other.

Example:

Students do not share ideas because they fear criticism.

Result:

People hide mistakes and avoid open communication.

2. Fear of Conflict

Meaning:

Team members avoid discussing disagreements.

Example:

Students stay silent even when they disagree with a decision.

Result:

Problems remain unsolved.

3. Lack of Commitment

Meaning:

The team does not clearly agree on decisions.

Example:

Students are unsure about the project plan.

Result:

Members feel confused and unmotivated.

4. Avoidance of Accountability

Meaning:

Team members do not hold each other responsible.

Example:

A student does not complete their task and no one says anything.

Result:

Work quality decreases.

5. Inattention to Results

Meaning:

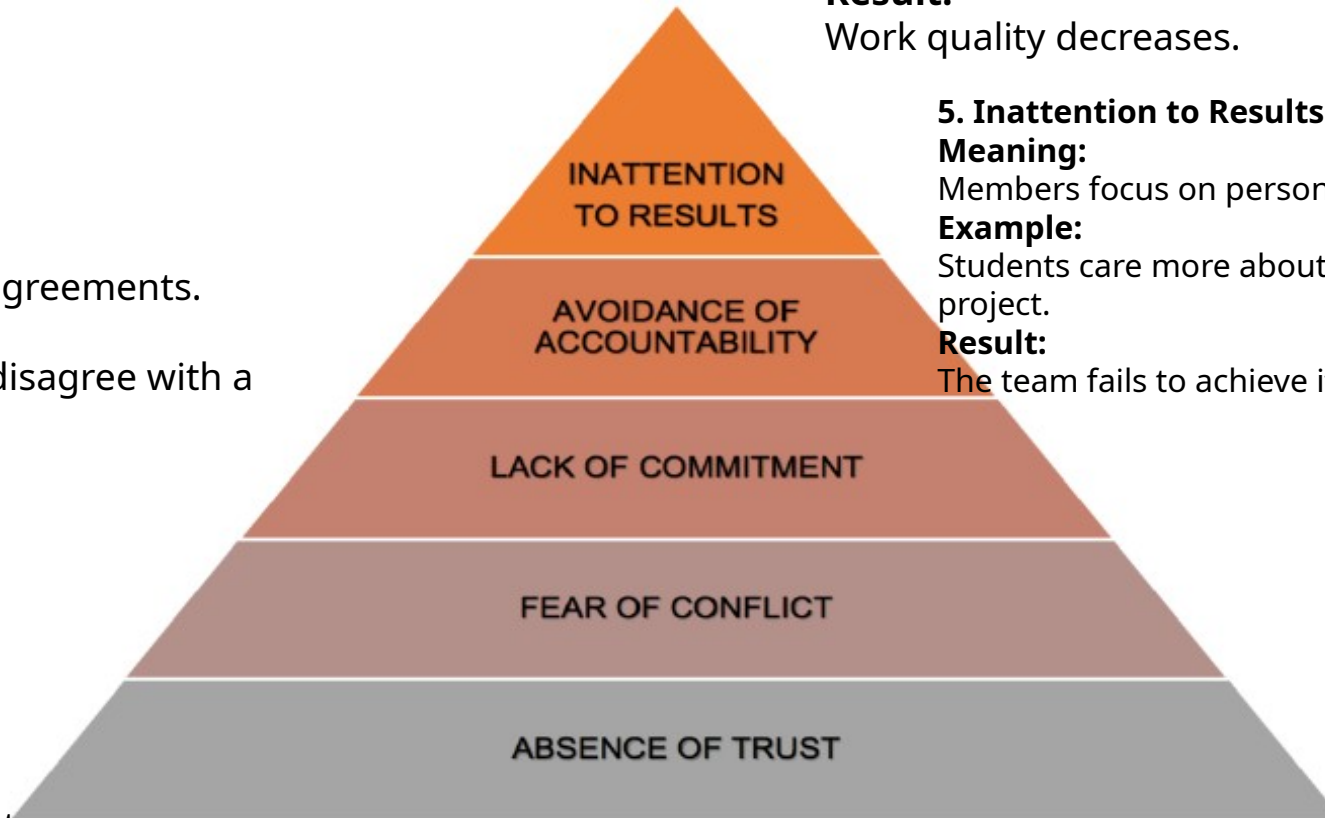
Members focus on personal success instead of team success.

Example:

Students care more about individual grades than the team project.

Result:

The team fails to achieve its goal.



Five Team Dysfunctions

- **Lack of Trust**

Team members **do not feel comfortable sharing ideas, mistakes, or asking for help.**

- **Fear of Conflict**

Teams **avoid discussion and disagreement**, which reduces **innovation and problem solving.**

- **Lack of Commitment**

Members **do not fully commit to decisions or tasks**, leading to **missed deadlines.**

- **Avoidance of Accountability**

Team members **do not hold each other responsible** for their work.

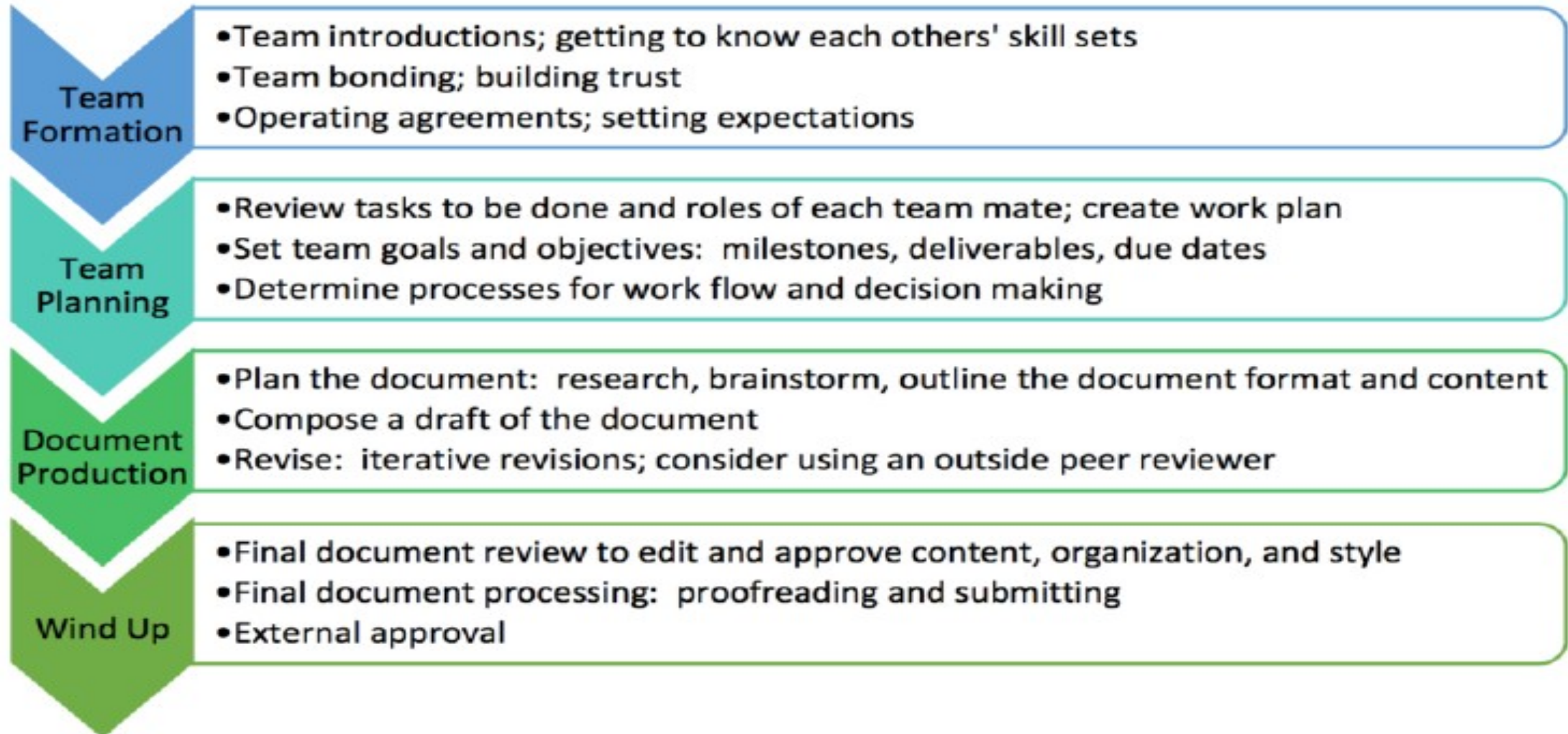
- **Inattention to Results**

Members focus on **personal goals instead of team or project success.**

Collaborative Writing

- Collaborative writing is when **multiple people work together to produce a document.**
- **In Engineering**
- Engineers collaborate to:
 - Gather information
 - Organize and manage data
 - Disseminate results
- The information is **analyzed to support important decisions**, so effective communication is essential.
- **Examples in Computer Science**
- Writing **software documentation**
- Preparing **project reports**
- Creating **research papers**
- Writing **system design documents**
- Preparing **presentations for software projects**

Collaborative Writing



Managing Team Conflict:

- **Team Conflict**
- Conflict can happen when people **work together in teams**.
- Some conflict can be **productive** and lead to **new ideas and innovation**.
- Poorly managed conflict can **harm the team and the project**.
- **Common Causes of Conflict**
- Unclear **team roles**
- Different **goals or expectations**
- Lack of **communication**

Strategies to Manage Team Conflict

1. **Respect team diversity** – understand different strengths and weaknesses.
2. **Avoid “silos”** – don’t split into “us vs them”; solve problems as a team.
3. **Address problems early** – communicate openly and review the **team charter** if needed.
4. **Don’t focus on minor issues** – compromise on small differences.
5. **Be persuasive, not aggressive** – explain your ideas respectfully.
6. **Separate people from the problem** – avoid blaming others.
7. **Focus on team interests** – aim for solutions that benefit the whole team.
8. **Understand team dynamics** – learning how teams work helps solve conflicts.



Thank
you